

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report evaluates the patent evidence provided for low acrylonitrile nitrile-butadiene rubber (NBR) synthesis and composition control. The target material is defined as NBR grades with reduced acrylonitrile content, typically emphasizing feed-ratio control, copolymer composition tuning, and polymerization conditions that govern final ACN level. In the supplied manifest, only one primary patent was identified: US10443339B2, which is not a direct NBR synthesis patent but is relevant to elastomeric materials used in harsh downhole environments where NBR and related rubbers are discussed as candidate seal materials. The patent focuses on reinforcing elastomeric matrices with carbon-based nanoplatelets at less than approximately 10 percent by weight to improve performance in high-temperature, high-pressure, and chemically aggressive service.

Because the evidence set contains no direct low-ACN NBR preparation example, no monomer feed ratios, initiator systems, emulsion recipes, or polymerization temperatures for NBR copolymerization are disclosed in the extracted text. Instead, the patent contributes indirectly by showing the application context for elastomer selection: packers, bore tools, and seals exposed to H₂S, CO₂, and HPHT conditions. The text explicitly contrasts HNBR and AFLAS with the target service environment, and it notes that elastomeric components may be formed by mixing elastomers with fillers and curing them into functional components.

Overall, the patent landscape here is sparse for direct low-acrylonitrile NBR synthesis. The available evidence supports the conclusion that low-ACN NBR remains a composition-controlled copolymer target rather than a process-described invention in this patent set. For synthesis-oriented work, the most relevant parameters to seek in additional patents would be butadiene/acrylonitrile feed ratio, ACN wt%, emulsion or solution polymerization conditions, and molecular-weight control strategy.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10443339B2
- Patent Title: Elastomeric materials
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose synthesis of low acrylonitrile NBR. It describes elastomeric components for bore tools and packers, where an elastomeric matrix is reinforced with carbon-based nanoplatelets and then cured or otherwise formed into downhole sealing elements. The text also discusses general elastomer mixing, reinforcement, and curing concepts rather than a specific NBR polymerization route. - Carbon-based nanoplatelets loading: less than approximately 10 percent by weight — Claim 1 - Mechanical property decrease temperature for HNBR: about 160 degrees C. — Detailed description - Mechanical property decrease temperature for AFLAS fluororubber: about 220 degrees C. — Detailed description - HPHT environment pressure: up to about 138 MPa — Detailed description - HPHT environment temperature: up to about 205 degrees C. — Detailed description - Ultra-HPHT environment pressure: up to about 241 MPa — Detailed description - Ultra-HPHT environment temperature: up to about 260 degrees C. — Detailed description - HPHT-hc environment pressure: greater than about 241 MPa — Detailed description - HPHT-hc environment temperature: greater than about 260 degrees C. — Detailed description - Young's modulus of ambient rubbers: of the order of about 3 MPa — Detailed description

Relevant Experimental Evidence - Example 1: No numbered experimental example was detected; the patent instead presents application concepts for elastomeric bore tools. - Example 2: The disclosure demonstrates that carbon-based nanoplatelets can be incorporated into an elastomeric matrix for downhole sealing components. - Example 3: The patent highlights use in packers, slips, and blowout protectors for harsh geologic environments. - Example 4: The text contrasts HNBR and AFLAS performance limits at elevated temperatures, motivating reinforced elastomer designs.

Technical Relevance This patent is only indirectly relevant to low acrylonitrile NBR because it identifies NBR-family elastomers as candidate materials for high-temperature, chemically aggressive oilfield service. It does not teach NBR copolymerization, but it provides a use-case context where low-ACN composition control and reinforcement strategies would matter.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The provided evidence does not include a direct low-acrylonitrile NBR synthesis patent. US10443339B2 is relevant mainly as an application-oriented elastomer reinforcement disclosure for harsh downhole service, not as a copolymerization method. Based on the evidence, the most defensible synthesis guidance for low-ACN NBR remains: control acrylonitrile content through monomer feed ratio, use a polymerization route that permits composition tuning, and consider reinforcement/filler strategies only after the base copolymer composition is established. No direct process parameters for NBR polymerization can be recommended from this patent set.

5. REFERENCES

- US10443339B2 | Elastomeric materials | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10443339B2/en>